

MSTAR PRACTICE WORKSHEET**Lesson 5-5: Determine the Volume of Rectangular Prisms**

Grade 6 Mathematics · Standard 6.GR.2

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 **SELECTED RESPONSE — CORRECT: C**What is the volume of a rectangular prism with $l = 7$ in, $w = 3$ in, $h = 4$ in?

- ☐ A 14 in³
- ☐ B 42 in³
- ☒ C 84 in³
- ☐ D 84 in²

$$V = l \times w \times h = 7 \times 3 \times 4 = 84 \text{ cubic inches.}$$

Question 2 **SELECTED RESPONSE — CORRECT: A**A time capsule box has a volume of 120 cm³. Its length is 10 cm and width is 4 cm. What is its height?

- ☒ A 3 cm
- ☐ B 4 cm
- ☐ C 6 cm
- ☐ D 12 cm

$$V = l \times w \times h \rightarrow 120 = 10 \times 4 \times h \rightarrow 120 = 40h \rightarrow h = 3 \text{ cm.}$$

Question 3 **SELECTED RESPONSE — CORRECT: B**

Which unit is used for volume?

- ☐ A Square inches (in^2)
- ☒ B Cubic inches (in^3)
- ☐ C Inches (in)
- ☐ D Degrees ($^\circ$)

Volume measures 3D space, so it uses cubic units like in^3 . Square units (in^2) are for area, and inches are for length.

Question 4 **SELECTED RESPONSE — CORRECT: A**

Two storage bins: Bin A is $8 \times 6 \times 5$ inches. Bin B is $9 \times 4 \times 7$ inches. Which holds more and by how much?

- ☒ A Bin B by 12 in^3
- ☐ B Bin A by 12 in^3
- ☐ C Bin B by 24 in^3
- ☐ D They hold the same

Bin A: $8 \times 6 \times 5 = 240 \text{ in}^3$. Bin B: $9 \times 4 \times 7 = 252 \text{ in}^3$. Bin B holds 12 in^3 more.

Question 5 **SELECTED RESPONSE — CORRECT: D**

A rectangular prism has length 8 cm, width 5 cm, and height 4 cm. What is its volume?

- ☐ A 17 cm^3
- ☐ B 80 cm^3
- ☐ C 120 cm^3
- ☒ D 160 cm^3

$V = l \times w \times h = 8 \times 5 \times 4 = 160 \text{ cm}^3$.

Question 6

SELECTED RESPONSE — CORRECT: B

A box has a volume of 72 in^3 . Its base is 6 in by 4 in. What is its height?

- ☐ A 2 in
- ☒ B 3 in
- ☐ C 12 in
- ☐ D 18 in

Height = $V \div (l \times w) = 72 \div (6 \times 4) = 72 \div 24 = 3 \text{ in}$.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7

TWO-PART QUESTION

Part A — correct answer: C

A rectangular time capsule container measures 8 inches long, 5 inches wide, and 3 inches tall. What is its volume?

- ☐ A 16 cubic inches
- ☐ B 40 cubic inches
- ☒ C 120 cubic inches
- ☐ D 158 cubic inches

$V = l \times w \times h = 8 \times 5 \times 3 = 120 \text{ cubic inches}$.

- **A:** That adds 8, 5 and 3. Volume multiplies the three measurements.
- **B:** That is 8 times 5, the base alone. Multiply by the height of 3 as well.
- **D:** That is the SURFACE AREA of the six faces, in square inches. Volume multiplies 8 by 5 by 3.

Part B — correct answer: B

Which expression shows the same volume written as base area times height?

- ☐ A $2 \times (8 \times 5)$
- ☒ B $(8 \times 5) \times 3$
- ☐ C $(8 + 5) \times 3$
- ☐ D $8 \times (5 + 3)$

The base is the 8 by 5 rectangle, so $B = 40 \text{ square inches}$, and $V = B \cdot h = 40 \times 3 = 120 \text{ cubic inches}$.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that correctly describe volume:

- ☒ **A** Volume is measured in cubic units.
- ☒ **B** $V = l \times w \times h$ for a rectangular prism.
- ☒ **C** Volume can also be found as base area $\times$ height.
- ☐ **D** Volume is measured in square units.
- ☒ **E** Doubling only the height of a prism doubles its volume.

A, B, C, and E are true. D is false — square units measure area; volume fills space and uses cubic units.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A box needs to hold exactly 60 cubic inches. Give three different sets of whole-number dimensions that work. Which set would make the box closest to a cube shape? Why might that matter?

Model answer: Any three whole numbers whose product is 60 work, such as $1 \times 6 \times 10$, $2 \times 5 \times 6$, and $3 \times 4 \times 5$. Of those, $3 \times 4 \times 5$ is closest to a cube because its edges are nearest to equal, and a cube-like box uses less material for its surface while holding the same 60 cubic inches.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.